Saheb Motiani

LLM interpretability, RL fine-tuning and ML systems

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Software engineer with 10+ years building production distributed systems at Meta, Babylon Health and Disney Streaming. I study LLM behavior with controlled experiments and build training and evaluation systems.

Selected Research

Independent work during an ML-focused career break · Aug 2025 – present

False self-correction in Llama-3.1-8B-Instruct

Writeup · Code and experiments

- Llama rejected correct arithmetic answers for some numbers (e.g. 100, 404) but not others. On **22 cases** where it rejected at baseline, three MLP activation edits cut rejection to **3/22**.
- Tested the edits with held-out unit selection and random controls; reverse patches also made selected, previously non-rejecting prompts produce rejection language.
- Tests of these two numbers suggested different early MLP components and a shared later influence on correctness judgments.

RL fine-tuning OLMo to withhold prohibited exam help

Writeup · Code and experiments

- Built a custom PyTorch/Transformers/PEFT GRPO trainer for OLMo-3-7B-Think-SFT with LoRA, vLLM rollouts and automated rewards; compared final-answer and whole-completion loss over 12 updates and 864 rollouts per run.
- Final-answer-only training increased rewarded answer tokens' relative likelihood in **47/51** eligible groups when saved reasoning was provided, yet fresh responses did not clearly improve.
- Whole-completion loss raised withholding from **68/144 to 81/144** fresh responses to seen prompts. Without policy text, it never refused prohibited help or asked permission (**0/24 each** on development prompts), so gains remained policy-dependent. One run per loss scope; automated judgments.

Professional Experience

Senior Software Engineer – Meta

Jan 2024 – May 2025

- Built notifications prompting non-admin users at high-value advertisers to appeal mistaken ad-account enforcements, unblocking **$6.6M/year** in revenue.
- Ran experiments showing broader appeal notifications increased harm, as bad actors exploited gaps in the appeal process. Worked with data scientists and the abuse-prevention team to limit the rollout to high-value advertisers.
- Fixed a multi-recipient concurrency bug, reducing notification failure rates by **70%** for Facebook Pages, **30%** for ad accounts and **8%** for businesses; found the platform logging false-positive delivery successes.

Senior Software Engineer – Babylon Health

Aug 2019 – Oct 2023

- Designed a Python/Faust pattern for deploying ML risk-prediction models, with outputs consumed by clinical portals and patient-outreach tools.
- Built reusable Flink patterns for ETL, feature engineering and stream processing; the data platform team began rolling out Flink internally.
- Led an **8–10-person cross-functional team** delivering a health-scoring product with Applied Scientists; developed Scala services for health-assessment and symptom-checker chatbot workflows.

Senior Software Engineer – Disney Streaming

Aug 2017 – Jun 2019

- Designed distributed services to validate incoming media before transcoding and packaging, connecting on-premises infrastructure and AWS through asynchronous Kafka communication.

Earlier Experience

Software Engineer – Cake Solutions (acquired by Disney)
Nov 2016 – Aug 2017 · Built media-acquisition control-plane automation for live ESPN+ and MLB.TV production.

Senior Associate – Morgan Stanley
Aug 2014 – Jul 2016 · Built SMS messaging infrastructure with third-party provider integrations.

Selected ML Systems & Experiments

Orbit Wars – self-play RL training and evaluation system
Kaggle placement: 1,060 / 4,729 · Code and results · Interactive replay

- Built a GPU-vectorized PyTorch simulator, entity-transformer PPO policy, self-play league and standalone export; measured **~4.1k end-to-end environment steps/s on RTX PRO 6000**, including policy inference and PPO updates.
- Diagnosed an evaluator that counted opponent crashes as wins; added fail-closed checks, built fixed, seat-balanced evaluation panels and checked train/export parity.
- A post-competition policy won **98.0%** against a mid-tier planner and **14.1%** against the 13th-ranked agent (256 games each), but **0/32** against a top-10 agent.

Snake – board-size generalization
Study and code · Demo

- An attention-pooling agent with egocentric inputs, trained on 10×10 and 20×20 boards, achieved **97.7% moves toward food** in zero-shot 100×100 play. Probes and training-size ablations pointed to training-state coverage as a bottleneck.

Technical Skills

ML and experiments: Python, PyTorch, JAX, NumPy, Transformers, PEFT/LoRA, vLLM; PPO, GRPO, self-play, activation patching, probing and controlled evaluation.

Production systems: Scala, Java, SQL, Kafka, Flink, AWS, GCP, Kubernetes; distributed services, streaming pipelines and model deployment.

Education & Open Source

B.Tech., Information and Communication Technology
Dhirubhai Ambani University (formerly DA-IICT) · 2014

Google Summer of Code (2013): ported the C++ Open Graph Drawing Framework to JavaScript with Emscripten.

Other Selected Projects

TasteBento · Poker RL · Content Moderation OpenEnv · JAX Transformer